

Two weaknesses of p_{quasar} : unranked quasars and top-end overconfidence

Internal note, `qso_pcolor`

4 October 2026

Summary

- **Unranked quasars.** Of 38,881 held-out test quasars above the faint limit, the scorer declines to rank 2.9% with model A (magnitude-dependent, `current`) and 3.2% with model B (magnitude-independent, `current_magindep`). For 89–92% of them the reason is *outside both models*: the colours are more than 4σ from both the quasar and the background model. The rest fail the QSO-support cut.
- **The cause is mainly the 4σ test, not the quasars.** The distance is a raw Mahalanobis distance in each object’s observed bands, and it grows with the number of bands k , while the cut is fixed. The unranked fraction is 0% for $k < 12$, 0.4–0.9% for $k = 13$ –15, 2–4% for $k = 17$ –19 and 7–15% for $k \geq 20$. The unranked quasars continue the distance distribution of the ranked ones just past the cut (Figs. 2, 3).
- Rescored with Legacy $g, r, z, W1, W2$ only, 93–94% of them are ranked; removing any one large survey (SDSS, PS1, AllWISE or the Legacy WISE bands) rescues 56–85%. No single survey or band is responsible. A secondary contribution is disagreement between surveys observed at different epochs (quasar variability): $|r_{\text{SDSS}} - r_{\text{Legacy}}| > 0.5 \text{ mag}$ for 33% of unranked quasars and 11% of ranked ones.
- **Fix (implemented, validated, the default since 5 October 2026):** calibrate the same distance statistic per object by simulating members of each class model in the object’s own bands with its own noise (Section 2.8). On the test panel, quasars outside both models fall from 2.5–2.9% to 0.14–0.17% and unranked quasars from 2.9–3.2% to 0.6%, with no dependence on the number of bands; background incidence at $p_{\text{quasar}} > 0.5$ changes by +0.05 percentage points. (A plain χ_k^2 conversion over-corrects: 2 of 38,881 quasars would remain flagged, because the distance is to the nearest of many mixture components.)
- **Top-end overconfidence is mostly a labelling effect.** In the natural test-cone population, objects with $p_{\text{quasar}} > 0.97$ are known quasars in 58–72% of cases (lower bound), but 171 of the 188 background objects there are already flagged as likely quasars (WISE AGN colours or Quaia). Counting those as quasars gives 0.93–0.99 (upper bound), close to the predicted values. Only 17 of the 188 are unflagged, 1.7–2.9% of all objects at $p_{\text{quasar}} > 0.97$ after weighting. The model’s top end is at most mildly overconfident; better labels, not a recalibration, would settle by how much.

1 Data and definitions

When the scorer declines to rank. After the faint limit (Legacy r $S/N \geq 10$), an object is not ranked if (i) it is *outside both models*, i.e. its Mahalanobis distance to the nearest component of the quasar model at any redshift, d_Q , and to the nearest component of the background model, d_B ,

both exceed `ood_flag_sigma = 4`; or (ii) it fails the *QSO-support cut*: the percentile of its colour density under the quasar model at the primary’s redshift is below 2/256. Both are computed in the object’s observed bands. Unranked objects get no p_{quasar} and no ranking statistic.

Samples.

- *Test panel* (Part 1): the main performance panel of the method note, $2 \times 20,000$ random held-out (role 3) quasars per hemisphere with their actual band coverage, of which 38,881 pass the faint limit. They were scored by both promoted bundles (the bundle files are identical to the promoted ones). That run used support threshold 6/256; we re-apply the promoted 2/256 from the stored percentiles.
- *Natural population* (Part 2): the calibration test of the method note (Section 10.6): all 1,640 known spectroscopic quasars in the 49 held-out test cones above the faint limit, and 40,000 of the cones’ background objects weighted by 2.75 to restore their natural share. The background sample still holds unidentified quasars; 1,443 of the 40,000 are flagged as likely quasars (Quaia membership, or Legacy $W1 - W2 > 0.8$ Vega).

Script: `scripts/method_unified/pquasar_diagnostics.py`; numbers in `pquasar_diagnostics.json`.

2 Part 1: real quasars left unranked

2.1 How many, and why

	A (magnitude-dependent)	B (magnitude-independent)
Test quasars	38,881	38,881
Unranked	1,113 (2.9%)	1,235 (3.2%)
outside both models	989	1,132
QSO-support cut	118	97
no quasar prior at this magnitude	6	6

806 quasars are unranked by both models, 307 by A only and 429 by B only. In the smaller natural sample of the method note the rates are 3.8% (A) and 3.5% (B), from about 60 objects.

2.2 Who they are

Figure 1 shows the unranked fraction against the quasars’ properties. It is flat in redshift (2.3–3.4% for $0.5 < z < 3.25$; 5% below $z = 0.5$) and rises to 7.5–15% above $z = 3.25$ (few quasars, where the quasar model is sparsest). In magnitude it is lowest at $r = 19$ –20 for model A (1.2%) and highest at $r = 21$ –22.5 (3.7–3.9%); model B is worse than A for bright quasars ($r < 19$: 2.8–5.5% against 1.3–2.8%), where its single colour width per redshift fits worst. It does not depend on extinction or Galactic latitude. The strongest trend is with the number of observed bands: 0.5–1.1% for 10–15 bands, 2.2% for 16–17, 3.9% for 18–19, 8.3% for 20–21, 10.5% for 22–23 and 20% for 24–25 (45 quasars).

P1. Fraction of held-out test quasars left unranked (38,881 quasars, Legacy r S/N ≥ 10 ; 68% intervals)

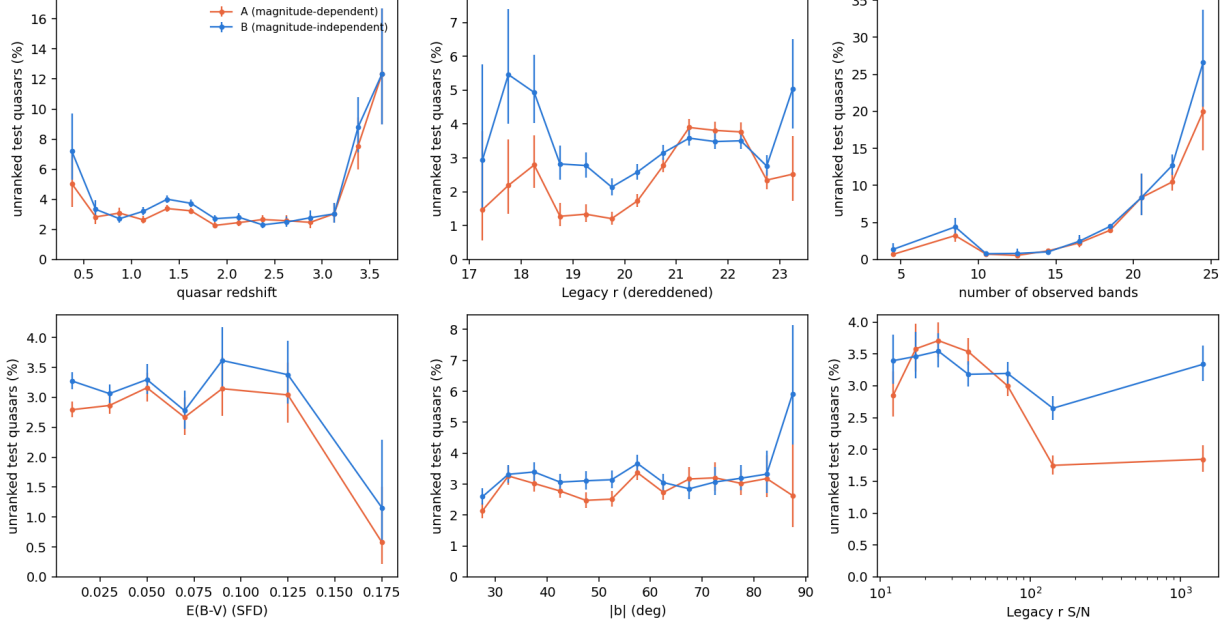


Figure 1: Fraction of held-out test quasars left unranked, against redshift, Legacy r , number of observed bands, $E(B-V)$, $|b|$ and Legacy r signal-to-noise, for models A (orange) and B (blue). Bars: 68% Wilson intervals; bins with fewer than 30 quasars are omitted.

2.3 They sit just past the cut

Figure 2 places every test quasar by its two distances. Ranked quasars fill a continuous cloud up to $d_Q = 4$; the outside-both-models quasars are its continuation, with d_Q mostly between 4 and 6. They are the tail of the same population, not a separate group of strange objects. Support-rejected quasars lie inside the cloud in d_Q : they are close to *some* quasar component, but not to the quasar model at their own redshift.

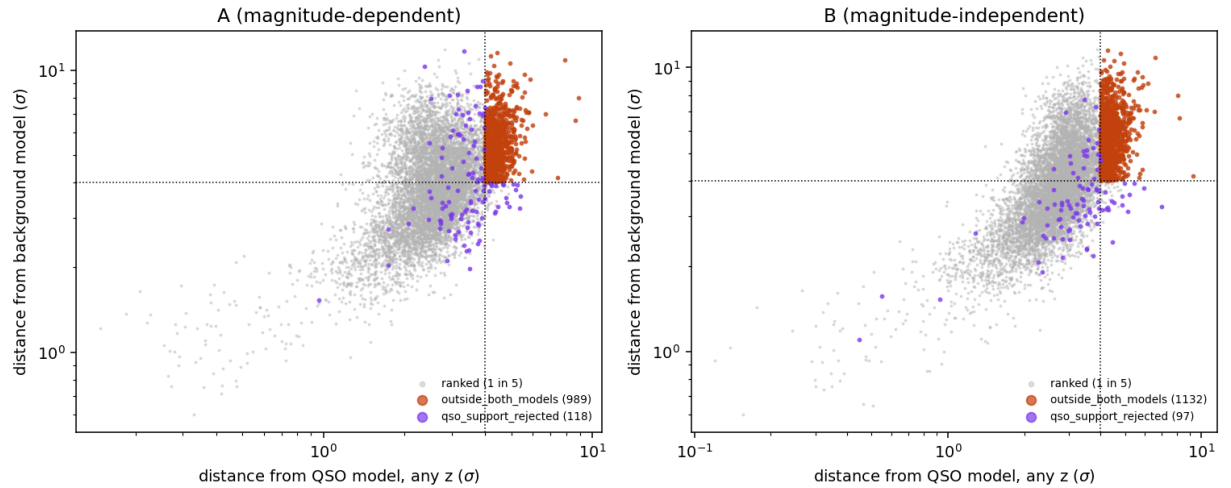


Figure 2: Distance from the quasar model at any redshift (d_Q) against distance from the background model (d_B), in σ , for test quasars: ranked (grey, one in five), outside both models (orange) and support-rejected (purple). Dotted: the 4σ cut on each.

2.4 The cause: the distance grows with the number of bands

For a Gaussian in k dimensions the squared Mahalanobis distance of a member follows χ_k^2 , whose median is about k ; a typical member lies at $d \approx \sqrt{k}$, i.e. 3.2σ at $k = 10$ and 4.5σ at $k = 20$. A fixed cut at 4σ is therefore strict for objects with many bands and loose for objects with few. The code documents this (the distance “is a different tail probability in 3 and in 4 dimensions”) but the cut does not correct for it. Figure 3 shows the effect in the data. The median d_Q of test quasars rises from 1.8 at $k = 10$ to 3.3 at $k = 23$, and its 99th percentile crosses 4 at $k \approx 14$. The fraction outside both models rises in step: 0% for $k \leq 12$, 0.4–0.9% for $k = 13$ –15, 2–4% for $k = 17$ –19, 7–10% for $k = 20$ –22 and 15% at $k = 23$. Background objects show the same rise, from at most 0.2% for $10 \leq k \leq 13$ to 16–18% at $k \geq 26$. The distance grows more slowly than $\sqrt{\chi_k^2}$ because it is measured to the nearest of many mixture components, so the textbook χ_k^2 correction overshoots: converted to an equivalent Gaussian significance with k degrees of freedom, only 2 (A) and 1 (B) of the 38,881 test quasars would remain outside both models.

P8. The raw Mahalanobis distance grows with the number of observed bands; the 4σ cut does not

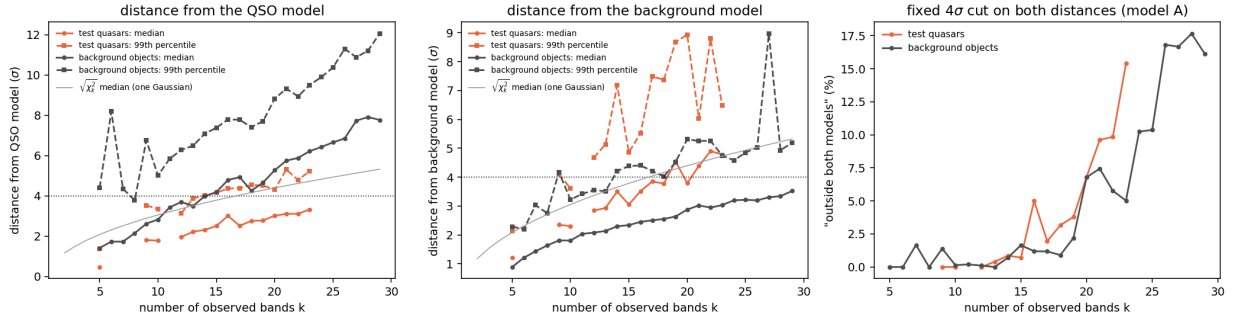


Figure 3: Left and middle: median and 99th percentile of the distance from the quasar model (d_Q) and from the background model (d_B) against the number of observed bands k , for test quasars (orange) and background objects (dark grey), model A. Grey line: median of $\sqrt{\chi_k^2}$ for a single Gaussian. Dotted: the 4σ cut. Right: fraction outside both models against k . Values of k with fewer than 30 objects are omitted.

2.5 Colours

Figure 4 overlays the unranked quasars (model A) on the colour distribution of the ranked ones in four redshift ranges. Most unranked quasars lie inside or at the edge of the ranked distribution in every Legacy colour plane; a minority are genuinely unusual (very red $g-r$ or $r-z$, which suggests dust reddening, or very blue $z-W1$). This agrees with the distance picture: in a few colours they look ordinary, and they are pushed past the cut by many small deviations across many bands.

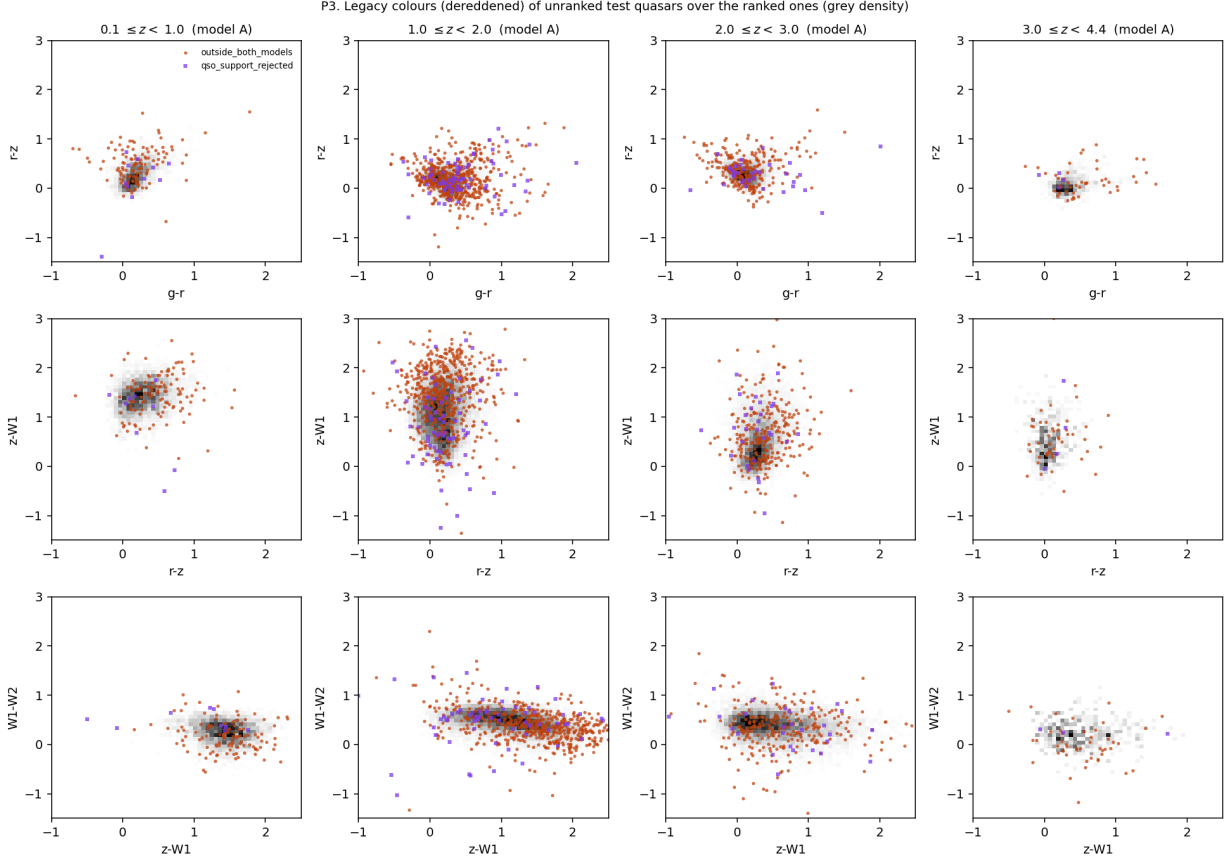


Figure 4: Legacy colours (dereddened luminosities) of unranked test quasars (orange: outside both models; purple: support-rejected) over the density of ranked test quasars (grey), in four redshift ranges, model A.

2.6 Bands and surveys

No single bad band. Under the quasar model at each quasar’s redshift (conditional on Legacy r), we computed the standardised residual of every observed band (Fig. 5). Only 1–3.5% of the unranked quasars’ bands deviate by more than 4σ , spread over SDSS, Legacy, AllWISE and PS1 with no dominant band; 84% (A) and 87% (B) of the unranked quasars have no band beyond 4σ at all (ranked: 99%). Their largest single residual is typically $2\text{--}3\sigma$, against $1\text{--}2\sigma$ for ranked quasars.

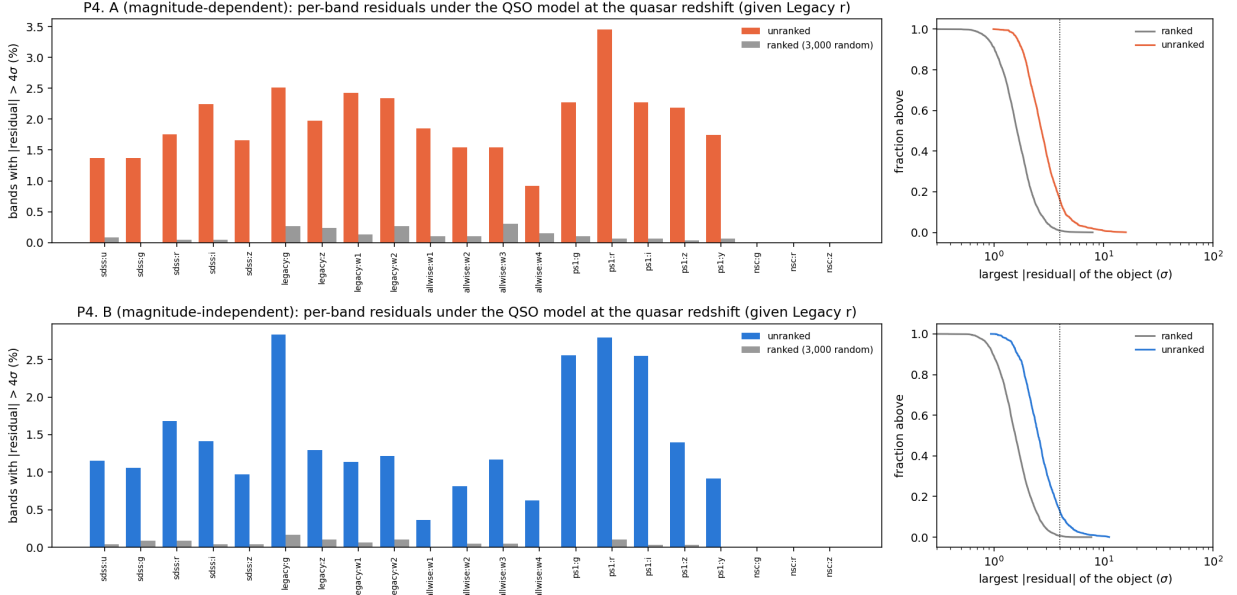


Figure 5: Left: percentage of observed bands with standardised residual $|R| > 4$ under the quasar model at the quasar’s redshift, given Legacy r , for unranked quasars and 3,000 random ranked ones, by band (Legacy north and south combined). Right: distribution of each object’s largest $|R|$. Top: model A; bottom: model B.

Survey-to-survey disagreement. The surveys observed at different epochs and calibrate differently. Figure 6 compares each quasar’s SDSS and PS1 r with its Legacy r . The unranked quasars have much broader differences: $|r_{\text{SDSS}} - r_{\text{Legacy}}| > 0.5 \text{ mag}$ for 33% of them (ranked: 11%), and $|r_{\text{PS1}} - r_{\text{Legacy}}| > 0.5 \text{ mag}$ for 28% (ranked: 9%). Quasar variability over the years between the surveys (SDSS imaging 2000–2008, PS1 2010–2014, Legacy DR9 2014–2019) is the natural explanation; the model treats the surveys as simultaneous. Unranked quasars also more often have SDSS (92% against 78%) and AllWISE (88% against 66%) photometry, i.e. more bands.

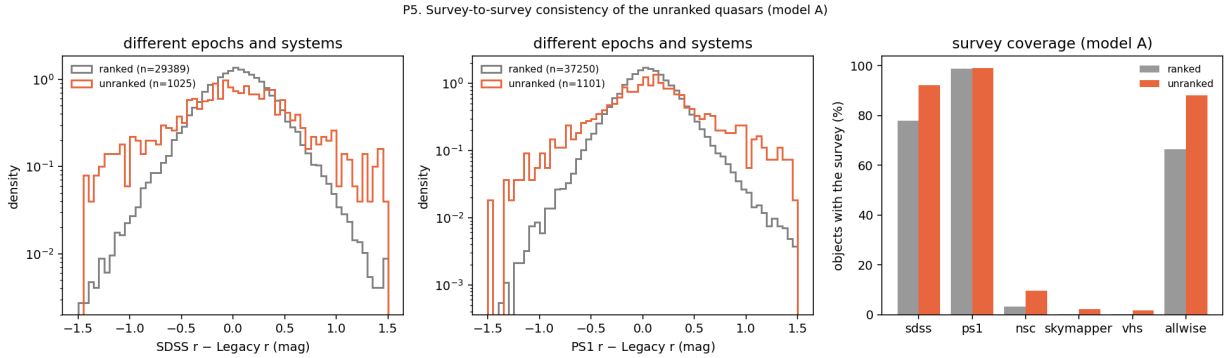


Figure 6: Left and middle: SDSS r and PS1 r minus Legacy r (dereddened luminosities) for ranked (grey) and unranked (orange) test quasars, model A. Right: fraction of each group with photometry from each survey.

Removing surveys. We rescored the unranked quasars of each model with Legacy bands only, and with each survey removed in turn (Fig. 7). Rescoring with all bands reproduces the original result (none ranked). With Legacy $g, r, z, W1, W2$ only, 93% (A) and 94% (B) are ranked. Removing SDSS ranks 78–79%, PS1 80–85%, AllWISE 70–75% and the Legacy WISE bands 56–57%; removing NSC, SkyMapper or VHS ranks 0.6–6%, because few of these quasars have them. That removing *any* large survey rescues most of them, rather than one specific survey, is the signature of the dimension effect, with variability between epochs adding to it.

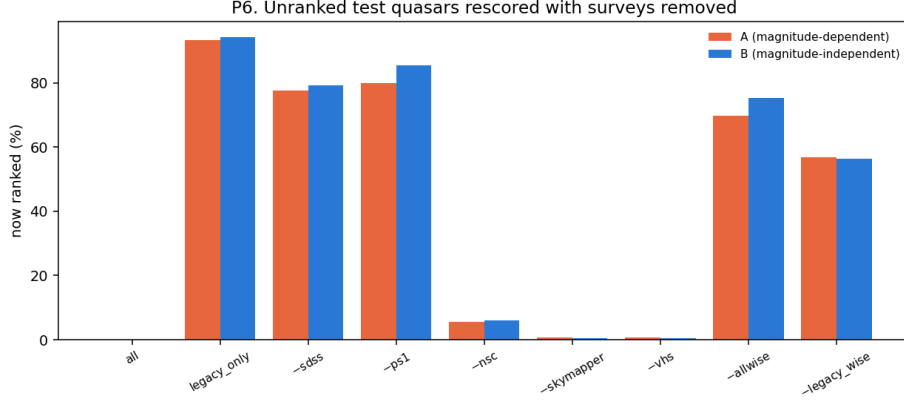


Figure 7: Unranked test quasars rescored with surveys removed: percentage now ranked, for models A and B. “all”: no band removed (reproduces the original); “legacy_only”: Legacy $g, r, z, W1, W2$ only; “-survey”: that survey’s bands removed.

2.7 What to do

1. **Make the outside-both-models threshold depend on the number of observed bands.** Calibrate $t(k)$ empirically on calibration-role objects (not the test set), for example so that a fixed small fraction (say 0.2%) of calibration quasars and of calibration background objects lie beyond it at each k , pooling sparse k . The χ_k^2 form is the wrong shape (it overshoots), but a scaled version may fit. Background objects cross the fixed cut in the same way (Fig. 3), so the change affects their abstention too; the AUC and the incidence of background objects at high p_{quasar} must be re-measured on the test set afterwards.
2. **Consider an inter-survey variability term:** an extra, magnitude- and time-baseline-dependent variance between surveys observed at different epochs, at least for quasars. This addresses the secondary cause and is a larger change to the model.
3. Until then: an unranked object with many bands is not evidence against a quasar. Rescoring it with Legacy bands only is a valid check; 93–94% of the unranked test quasars are ranked that way.

2.8 The fix: a calibrated outside-both-models test

Implemented in `qso_pcolor.oood_calibration` and available in the scorer as `ood_calibration=dict(alpha, draws, seed)`; the default of `UnifiedPSFModel.score` since 5 October 2026. The statistic is unchanged: the squared nearest-component distance T_c of the non-reference bands given Legacy r , with the object’s noise added, for the quasar model (all redshift slices, equal weight) and for the background model (weights at the object’s sky position). For each object we draw 256 members of each class model conditioned on the object’s own r , in its own observed bands and with its own noise, compute the same statistic, and form

$$p_c = \frac{1 + \#\{T_c^{\text{fake}} \geq T_c^{\text{obs}}\}}{257}.$$

The object is outside both models when $p_Q < \alpha$ and $p_B < \alpha$, with $\alpha = 2/256$ set in advance (as for the support cut). This is a parametric-bootstrap tail probability of the existing statistic, so it handles the number and the identity of the observed bands, the noise and the nearest-of-many-components effect at once. It costs 13ms per object for both models. Unit tests check that p is uniform for members of a known mixture in 4, 12 and 25 dimensions (where the raw distance of a typical member exceeds 3σ), and that the scorer path reproduces the offline values and is independent of batch order.

Calibration role (a-priori check of α). On 4,000 calibration-role quasars and 4,000 background objects, the new test flags 0.23–0.25% of quasars and 0.48–0.60% of background objects; a class member falls below α in its own model in 0.4% (quasars) and 0.75% (background) of cases, against the nominal 0.78%.

Test panel. Figure 8 and the table compare the two rules on the held-out panel (38,881 quasars, 14,247 background objects). The objects whose flag clears were rescored through the scorer.

	A (magnitude-dependent)		B (magnitude-independent)	
	fixed 4σ	calibrated	fixed 4σ	calibrated
Quasars outside both models	2.54%	0.14%	2.91%	0.17%
Quasars unranked (all reasons)	2.86%	0.61%	3.18%	0.57%
Background outside both models	2.09%	0.51%	2.14%	0.53%
Background unranked (all reasons)	69.2%	69.1%	59.9%	59.7%
Quasar recall ($p_{\text{quasar}} > 0.5$)	0.891	0.913	0.859	0.883
Background incidence ($p_{\text{quasar}} > 0.5$)	1.14%	1.19%	1.08%	1.14%
AUC ($\log \mathcal{R}$, unranked at the bottom)	0.980	0.994	0.976	0.994

The refusal rate no longer depends on the number of bands, for quasars or background objects. Of the 947 (A) and 1,079 (B) quasars whose flag clears, 93–95% are then ranked; the rest fail the support cut. Only 7 (A) and 9 (B) quasars are newly flagged. Almost all background objects whose flag clears (245 of 263, A) are still not ranked, because the support cut catches them, so background incidence rises by only 0.05–0.06 percentage points (7–8 of 14,247 objects). The AUC rises by 0.014–0.018 because the AUC as defined counts an unranked quasar as ranked below every ranked background object and tied with the unranked ones: moving 2.3–2.6% of quasars from the bottom to their scores accounts for the change. The real quasars’ p_Q values lie below the uniform line (Fig. 8, right): the quasar model is, if anything, slightly broad, so the test is conservative.

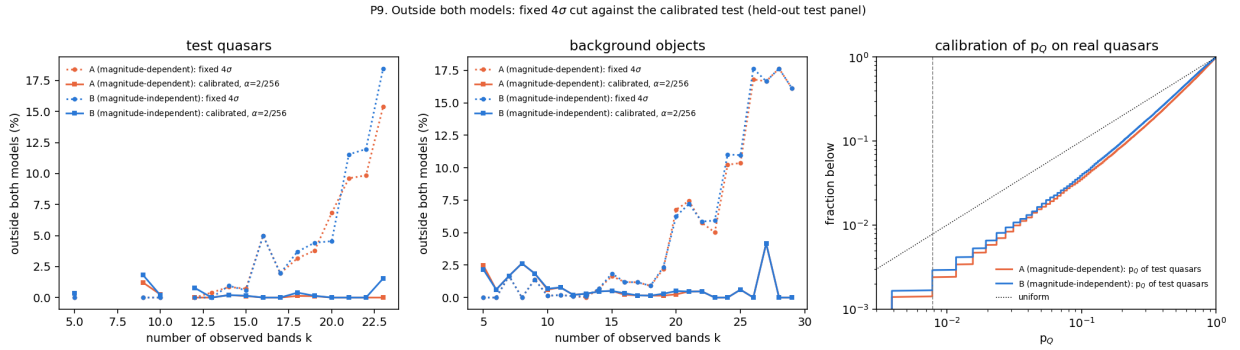


Figure 8: Outside-both-models fraction against the number of observed bands for the fixed 4σ cut (dotted) and the calibrated test with $\alpha = 2/256$ (solid), for test quasars (left) and background objects (middle), models A and B. Right: cumulative distribution of p_Q for the test quasars (dotted: uniform; dashed: α).

3 Part 2: overconfidence at the top

3.1 Reliability above $p_{\text{quasar}} = 0.5$

Among ranked objects of the natural population we bin p_{quasar} and count known quasars, flagged background objects and unflagged background objects (each background object weighted by 2.75). The observed quasar fraction is bracketed by a lower bound (only known quasars count) and an upper bound (flagged background objects also count).

p_{quasar} bin	mean p_{quasar}	known QSOs	flagged bkg	unflagged bkg	lower	upper
<i>Model A</i>						
0.90–0.95	0.926	82	28	6	0.47	0.91
0.95–0.97	0.961	42	13	3	0.49	0.90
0.97–0.99	0.982	122	28	4	0.58	0.95
0.99–0.999	0.996	213	42	8	0.61	0.94
> 0.999	1.000	755	101	5	0.72	0.99
<i>Model B</i>						
0.90–0.95	0.929	74	23	3	0.51	0.94
0.95–0.97	0.961	48	13	2	0.54	0.94
0.97–0.99	0.982	127	21	5	0.64	0.93
0.99–0.999	0.996	206	36	6	0.64	0.95
> 0.999	1.000	704	102	6	0.70	0.98

Background counts are unweighted; the bounds use the weight. At $p_{\text{quasar}} > 0.9$ the upper bound is 0.01–0.06 below the mean prediction, and the lower bound 0.28–0.47 below it (Fig. 9, left).

3.2 Who the high- p_{quasar} background objects are

With model A, 188 background objects have $p_{\text{quasar}} > 0.97$. 171 (91%) are flagged as likely quasars: 118 by WISE colour alone, 36 by WISE colour with Gaia astrometry consistent with zero proper motion, and 17 by Quasia membership (all also WISE-flagged). Only 17 are unflagged (16 with no flag of any kind, one with only the zero-proper-motion diagnostic, which is not part of the flag). Their magnitudes are those of the high- p_{quasar} known quasars (median $r = 21.2$ against 20.7), and in $W1 - W2$ they lie with the known quasars, above the stellar locus (Fig. 9, middle and right). The WISE flag requires $S/N \geq 5$ in both $W1$ and $W2$. Of the 17 unflagged objects, 3 have $W1 - W2$ above the AGN threshold but $W2 S/N < 5$, 4 more have too little $W1$ or $W2$ signal for a reliable colour, and 10 have stellar-like $W1 - W2$ (-0.3 to $+0.13$ AB, several close to the threshold of 0.16) with good S/N . These 10 are the plausible genuine contaminants. After weighting, all 17 unflagged objects are 2.9% of the objects at $p_{\text{quasar}} > 0.97$, and the 10 with stellar infrared colours 1.7%.

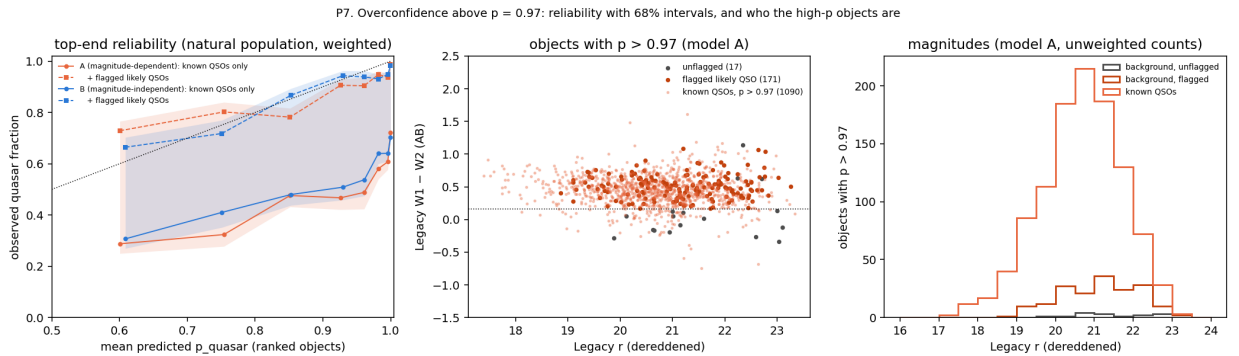


Figure 9: Left: observed quasar fraction against mean predicted p_{quasar} for ranked objects of the natural test-cone population, counting known quasars only (solid, lower bound) or also flagged likely quasars (dashed, upper bound); shading: 68% Wilson intervals of the two bounds. Middle: Legacy $W1 - W2$ (AB) against r for objects with $p_{\text{quasar}} > 0.97$ under model A: known quasars (light), flagged background objects (dark orange) and unflagged ones (black). Dotted: the WISE AGN threshold ($W1 - W2 = 0.8$ Vega). Right: their r distributions (unweighted).

3.3 Interpretation

The apparent overconfidence of the method note (“ $p_{\text{quasar}} > 0.99$ gives 0.69–0.98 observed”) is mostly a property of the labels. The background sample is a photometric catalogue with known quasars removed; it still contains quasars that have no spectrum, and the objects the model is

most confident about are exactly those. If the flagged objects are quasars, the top end is calibrated to within a few per cent; the unflagged objects bound the genuine contamination at 1.7–2.9% at $p_{\text{quasar}} > 0.97$ (more if some flagged objects are not quasars). A recalibration fitted to the lower-bound labels would make the model *underconfident* on real quasars, which is why none was applied. The ranking is unaffected either way.

What would settle it. Spectroscopic labels for the flagged objects: cross-matching them with DESI DR1 spectra not used as training labels, or a newer release, would move the lower bound towards the upper. With 171 high- p_{quasar} flagged objects in the test cones, a few dozen spectra would already measure the fraction.

4 Conclusions

- The 2.9–3.2% of real quasars left unranked are mainly an artefact of a fixed 4σ cut on a distance that grows with the number of observed bands. They are ordinary quasars with many bands, plus a smaller contribution from variability between surveys. The calibrated test (Section 2.8) recovers 80% of them without raising background incidence, and is now the default.
- The top-end overconfidence is mostly unlabelled quasars in the background sample. The model’s highest probabilities are consistent with the upper label bound; unflagged background objects make up 1.7–2.9% of the objects at $p_{\text{quasar}} > 0.97$.